

OBJECT ORIENTED PROGRAMMING

#4

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EXERCISE #1

```
#include <iostream>
```

```
using namespace std;
```

```
class Point {
```

```
private:
```

```
    double x;
```

```
    double y;
```

```
public:
```

```
    Point(double xcord, double ycord) {
```

```
        x = xcord;
```

```
        y = ycord;
```

```
    }
```

```
    void info() {
```

```
        cout << "(x,y) = " << x << ", " << y << endl;
```

```
    }
```

```
    double getX() {
```

```
        return x;
```

```
    }
```

```
    double getY() {
```

```
        return y;
```

```
    }
```

```
    void getP(double& xcord, double& ycord) {
```

```
        xcord = x;
```

```
        ycord = y;
```

```
    }
```

```
};
```

```
int main(void) {
```

```
    double x, y;
```

```
    Point p {10.5, 20.99};
```

```

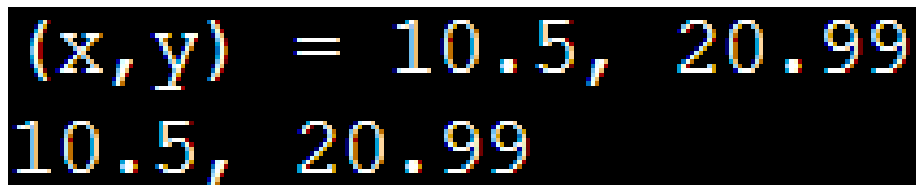
p.info();
p.getP(x, y);

cout << x << ", " << y << endl;

return 0;
}

```

- **This code uses class. The class has member variables and a member function inside it. This code prints the data that has been assigned to a private member of a class named 'Point'.**



```

(x, y) = 10.5, 20.99
10.5, 20.99

```

EXERCISE #5

```

#include <iostream>

using namespace std;

class Triangle {
    double m_base;
    double m_height;

public:
    Triangle(double a, double b) {
        m_base = a;
        m_height = b;
    }

    double getBase() const {
        return m_base;
    }
}

```

```

double getHeight() const {
    return m_height;
}

double area() const {
    double area = m_base * m_height / 2;
    return area;
}

~Triangle() {
    cout << "Destructor: base " << m_base << ", height " << m_height << "." << endl;
}

};

int main() {
    Triangle t1{ 10.5, 4.5 };
    const Triangle t2{ 7.0, 3.0 };

    cout << "Triangle t1{ 10.5, 4.5 }" << endl;
    cout << "    base: " << t1.getBase() << endl;
    cout << "    height: " << t1.getHeight() << endl;
    cout << "    area: " << t1.area() << endl << endl;

    cout << "const Triangle t2{ 7.0, 3.0 }" << endl;
    cout << "    base: " << t2.getBase() << endl;
    cout << "    height: " << t2.getHeight() << endl;
    cout << "    area: " << t2.area() << endl << endl;

    return 0;
}

```

- **This code uses class too. The class has member variables and a member function inside it. This code prints the data that has been assigned to a private member. The**

unique characteristic of this code is that it has constant data.

```
Triangle t1{ 10.5, 4.5 }  
    base: 10.5  
    height: 4.5  
    area: 23.625  
  
const Triangle t2{ 7.0, 3.0 }  
    base: 7  
    height: 3  
    area: 10.5  
  
Destructor: base 7, height 3 .  
Destructor: base 10.5, height 4.5 .
```